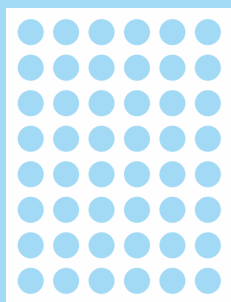




Certificate of Analysis



Olink[®] Explore

PROJECT NAME

ISSUE DATE

8/29/2022

1. Project information

No. of samples	No. of plates	Normalization method
684	8	Intensity normalization

1.1 Sample type

N/A

1.2 Project specific comments

N/A

2. Quality control

Three internal controls are added to each sample, the Incubation control, the Extension Control and the Amplification control. The Extension Control is used for the generation of the NPX values. The Incubation Control and the Amplification Control are used to monitor the quality of assay performance, as well as the quality of individual samples.

Three external controls are included in each run, the Plate Control (healthy pooled plasma), Sample Control (healthy pooled plasma) and Negative Control. The Plate Control is used for data normalization, the Sample Control is used to assess potential variation between runs and plates, and the Negative Control is used to calculate Limit of Detection for each assay and to assess potential contamination of assays.

The following parameters are evaluated in the Quality Control(QC) :

1. The average matched counts¹ for each sample. To pass QC, there should be at least 500 counts, otherwise the sample receives a QC warning status.
2. The deviation from the median value of the Incubation- and Amplification Controls for each individual sample. To pass QC, the deviation should not exceed ± 0.3 NPX for either of the internal controls, otherwise the sample will receive a QC warning status.
3. The deviation of the median value of the Negative Controls from a predefined value set for each assay. To pass QC, the deviation of the median of the Negative Controls must be ≤ 5 standard deviations from the set predefined value, otherwise the assay will receive a warning status.

¹The number of reads for each specific combination of sample and assay

All samples included in the project are presented in the data output file. Samples that do not pass the QC are indicated with WARN in the column named QC_warning. Data points from samples that do not pass QC should be treated with caution. Manual QC warnings are indicated with MANUAL_WARN in the column named QC_warning. Section 2.1 reports the fraction of samples that pass QC for all assays per panel and the fraction of data points passing QC per panel. Samples with manual QC warning are counted as not passed QC. Assays that do not pass the QC are indicated with WARN in the column named Assay_warning. Data points from assays that do not pass QC should be treated with caution.

2.1 QC summary

Olink Panel	Samples passed QC	Samples passed QC (%)	Datapoints passed QC	Datapoints passed QC (%)
Explore 384 Cardiometabolic	626 / 684	92	246692 / 252396	98
Explore 384 Inflammation	629 / 684	92	246369 / 251712	98
Explore 384 Neurology	605 / 684	88	242540 / 251028	97
Explore 384 Oncology	598 / 684	87	242154 / 251712	96

2.2 Intra - and Inter - assay Coefficient of Variance(% CV)

Intra - and inter - CVs are based on the Sample Controls(pooled plasma samples) included on each sample plate. Calculations are made for each assay using NPX - values.Average % CV for all assays on a panel is presented in section 2.2.1.The number of assays with CVs within defined intervals are presented in section 2.2.2.

2.2.1 Average %CV

Olink Panel	Intra-assay %CV	Inter-assay %CV
Explore 384 Cardiometabolic	8	24
Explore 384 Inflammation	9	24
Explore 384 Neurology	10	22
Explore 384 Oncology	11	21

2.2.2 Intra-assay %CV distribution

Olink Panel	<5%	≥5-<10%	≥10-<15%	≥15%	N/A*
Explore 384 Cardiometabolic	77	193	54	28	17
Explore 384 Inflammation	84	164	41	35	44
Explore 384 Neurology	27	170	68	49	53
Explore 384 Oncology	69	102	75	90	32

*Assays where CV is not possible to calculate

2.2.3 Inter-assay %CV distribution

Olink Panel	<10%	≥10-<20%	≥20-<30%	≥30%	N/A*
Explore 384 Cardiometabolic	5	143	150	58	13
Explore 384 Inflammation	10	103	163	60	32
Explore 384 Neurology	8	159	124	41	35
Explore 384 Oncology	16	174	112	46	20

*Assays where CV is not possible to calculate

3. Protein detection results

3.1 Number of proteins detected in >50% of the samples

Olink Panel	No. of detected proteins / Total no. of proteins	Detected proteins (%)	Expected detectability in EDTA plasma* (%)
Explore 384 Cardiometabolic	293 / 369	79	N/A
Explore 384 Inflammation	196 / 368	53	N/A
Explore 384 Neurology	225 / 367	61	N/A
Explore 384 Oncology	183 / 368	50	N/A

*The expected detectability is based on EDTA plasma from healthy donors. These values are intended as guidelines only and protein levels may vary depending on different pathological conditions, sample matrices, or sample preparation methods.

3.2 Data output

Data is presented as NPX (Normalized Protein eXpression) values. NPX is Olink's relative protein quantification unit on log2 scale. NPX values are calculated from the number of matched counts, using NGS (Next Generation Sequencing) as readout. Data values for measurements below limit of detection (LOD) are reported for all samples.